

SURAJ VINOD MISKIN

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PROFESSIONAL SUMMARY

Result-oriented **Embedded Engineer** with a strong foundation in automotive electronics and **ADAS (Advanced Driver Assistance Systems)**. Hands-on experience in ECU development, automotive networking, and microcontroller-based system design. Proficient in **C, C++, and Python**, with practical exposure to **Vector tools** and real-world automotive projects during tenure at Tata Elxsi, contributing to client deliverables for Mahindra & Mahindra.

EXPERIENCE

Engineer – ADAS System Digital Validation <i>TATA Elxsi (Onsite at Mercedes-Benz R&D)</i>	Nov 2025 – Present <i>Bengaluru, India</i>
- Developed and deployed Python-based automation frameworks to streamline ADAS ECU validation workflows, reducing manual analysis effort and improving execution efficiency. - Designed and implemented a GUI-based automation tool (CustomTkinter) for end-to-end test execution, including test case selection and automated report triggering. - Automated Pcap to DAT conversion pipeline by building parsing and formatting scripts to process large Ethernet network traces for ADAS signal validation. - Integrated automation with Linux-based test benches using SSH-based remote command execution to trigger evaluation scripts. - Collaborated with cross-functional teams to debug integration issues and deliver high-quality validation reports to OEM stakeholders.	
Engineer (Project: Mahindra & Mahindra) <i>TATA Elxsi (Onsite Client Location)</i>	Dec 2024 – Oct 2025 <i>Chakan, Pune</i>
- Served as a key member of the DTC (Diagnostic Trouble Code) Clearance Team, ensuring vehicle system integrity and functional safety before production release. - Utilized the Mahindra Intelligence Diagnostic Assistance (MIDA) tool to identify, diagnose, and clear vehicle faults across multiple engine and body control modules. - Gained hands-on expertise in CAN communication and diagnostic validation workflows, working directly on vehicle ECUs to resolve integration issues. - Coordinated with production and quality teams to validate fix implementations, ensuring zero-defect rollout for upcoming vehicle models.	

PROJECTS

Python GUI Automation Tool <i>CustomTkinter, Python, Pcap-to-DAT</i>	2025
- Architected a GUI-based tool to automate the conversion of network traffic (Pcap) files into DAT format. - Reduced manual data processing time significantly by implementing automated file parsing and formatting logic.	
Embedded Power Window System <i>Embedded C, PIC16F877A</i>	2025
- Developed firmware for a microcontroller-based power window system featuring anti-pinch safety logic. - Simulated and tested real-time response times for dual-mode manual and automatic operations.	

EDUCATION

KLE Technological University <i>Bachelor of Engineering in Electrical and Electronics; GPA: 8.29</i>	Karnataka, India Sep 2021 – May 2024
Relevant Coursework: Data Structures, OOP concepts with C++, Machine Learning, OS & Embedded Systems, EV Technologies, BMS.	

TECHNICAL SKILLS

Automotive Tools: Vector CANoe, MIDA, HIL Testing
Languages & Protocols: Python (Intermediate), CAPL, Embedded C, UDS, CAN, Ethernet
Embedded Systems: ADAS Level 2, IDC ECU Testing, PIC16F877A, DTC Clearance
Tools: Git, VS Code, Linux, CustomTkinter, Docker

PUBLICATIONS

Intrusion Detection System for Electric Vehicle Charging Station | *IEEE Xplore*

Feb 2024

- Published research in the 2023 3rd International Conference on Mobile Networks and Wireless Communications (ICMNWC).
- Developed a Deep Learning-based IDS using TensorFlow to detect and prevent cybersecurity threats in EV infrastructure.

EXTRA CURRICULAR ACTIVITIES

Collaborated with the **Anaconda Navigator UI/UX design team**, sharing structured feedback and usability insights to enhance interface design and workflow efficiency.

Recognized for constructive reviews and continuous engagement in improving product usability and developer experience.